A Bias Correction Semi-Supervised Semantic Segmentation Framework for Remote Sensing Images

Li Zhang, Zhenshan Tan, Yuzhi Zheng, Guo Zhang, Wen Zhang, and Zhijiang Li

***Abstract*— The clustering assumption is widely adopted in semi-supervised semantic segmentation methods. However, this assumption heavily relies on high-quality feature representation, leading to learning and cognitive biases if it does not hold. Learning bias entails the potential overfitting of labeled data, leading to capturing local features and consequently misclassifying semantic categories. Cognitive bias indicates the network's susceptibility to interference features. Recently, a consistency-based mechanism has been proposed to address these biases. By subjecting unlabeled data to diverse weak perturbations, it breaks original clustering features, compelling the model to learn more robust and generalized representations. However, when processing complex remote sensing images, these weak perturbations often prove ineffective in the later stages of training. To address this, we propose a bias correction framework (BCF). The BCF begins with a feature consistency enhancement module (CEM) that guides the student model to learn feature representations with greater generalization capability. In the feature decoding stage, we introduce a multidecoder structure with weakly orthogonal weights to maximize feature differences, thereby further reducing learning and cognitive biases. In addition, to improve the confidence of pseudolabels and enhance consistency learning, we design a multidecoder teacher model based on symmetrical knowledge transfer, allowing the diverse and multiangle information learned by the student model to be transferred to the teacher model. Extensive experimental results show that our method significantly outperforms state-of-the-art methods on the ISPRS Potsdam and Vaihingen datasets.**

***Index Terms*— Consistency-based learning, feature perturbations, remote sensing images, semi-supervised semantic segmentation, weak orthogonality constraint.**

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Li Zhang and Guo Zhang are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: zhangli0826@whu.edu.cn; guozhang@whu.edu.cn).

Zhenshan Tan is with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: zstan@nuist.edu.cn).

Yuzhi Zheng is with the School of Law, Wuhan University, Wuhan 430072, China (e-mail: yuzhi_zheng@whu.edu.cn).

Wen Zhang is with the School of Information Management, Wuhan University, Wuhan 430079, China (e-mail: wenzhang.vcl@whu.edu.cn).

Zhijiang Li is with the School of Information Management, Wuhan University, Wuhan 430079, China, and also with the Luojia Laboratory, Wuhan 430075, China (e-mail: lizhijiang@whu.edu.cn).

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I. Introduction

FULLY supervised semantic segmentation methods represented by FCN, U-Net, and Deeplab. have achieved impressive performance [1], [2], [3], [4], [5]. However, they require a large amount of annotated data and pixel-level manual annotation by individuals with professional knowledge, which are expensive and time-consuming [6], [7]. Compared with expensive annotated data, unannotated remote sensing image data are much easier to obtain. Therefore, semi-supervised remote sensing image semantic segmentation methods are proposed to make full use of relatively small amounts of labeled data and combine them with a large amount of unlabeled data [8], [9], [10].

Most semi-supervised semantic segmentation methods are based on an important assumption, i.e., the clustering assumption, which posits the existence of a cluster structure in unlabeled data, where similar samples are close to each other in the feature space [11]. However, this assumption heavily relies on high-quality feature representations. In the context of complex remote sensing images, which often have low-quality feature representations, this assumption can introduce significant learning and cognitive biases. On the one hand, the clearer supervision signal in the clustering feature space causes the model to overfit the labeled data, limiting the effectiveness of the features learned from the unlabeled data. As training progresses, this limitation becomes more pronounced, reducing the complementarity between the labeled and unlabeled data and leading to a learning bias [12]. This bias results in the network favoring certain patterns, potentially producing incorrect classification outcomes. On the other hand, when two categories are highly similar in the feature space, the clustering algorithm may misclassify their feature points, further constraining the network's cognitive ability. This issue introduces cognitive bias, which reduces the model's capacity to differentiate between these similar categories. If the model generates incorrect pseudolabels for unlabeled data of similar categories, these inaccuracies can reinforce the model's misunderstanding, exacerbating the cognitive bias.

To address the above problems, consistency-based semi-supervised semantic segmentation methods are adopted [12], [13], [14]. The core idea of these methods is to modify the feature representations of data by applying various perturbations to the unlabeled data, thereby disrupting the clustering

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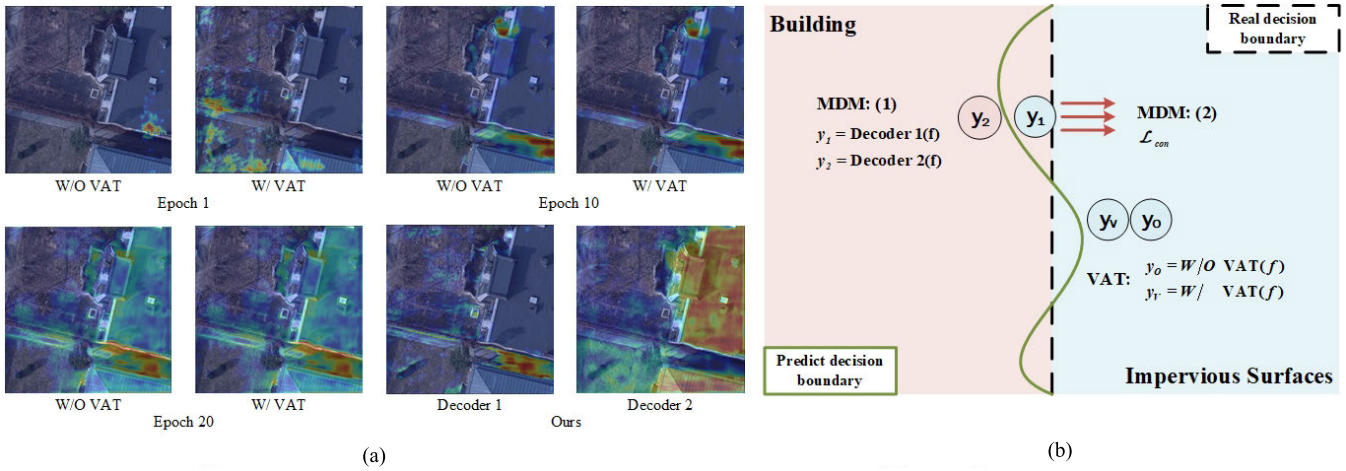


Fig. 1. Visual comparison between MDM and VAT. (a) Feature visualization With CAM: from Epochs 1 to 20 are the CAM figures of the baseline network before and after adding VAT during training. The last one is the CAM figure of the baseline network with an MDM added. (b) Light blue represents "building" and light pink represents "IS." The red arrow indicates the direction in which the decision boundary moves.

structure and compelling the model to learn more robust and generalized representations [13]. Importantly, to preserve the original data distribution, existing methods [14], [15], [16], [17], [18], [19] typically apply weak noise or slight transformations to the features. However, we are surprised to find that these weak perturbations do not always yield the expected results. As shown in Fig. 1, we use an image from the Potsdam dataset and take "building" as an example. The data, platform, and parameters used in all experiments in this article are kept consistent during training. For more details, see Sections IV-A and IV-B. We use a class activation map (CAM) [20] to perform a simple feature visualization. A typical feature perturbation method virtual adversarial training (VAT) [21] is added to the consistency-based semi-supervised semantic segmentation network. As depicted in Fig. 1(a), in the early stage of training (Epoch 1), weak feature perturbations cause the network to focus on different areas. However, as the training progresses (Epoch 10 and Epoch 20), the descent of the loss function tends toward a plateau, accompanied by a diminishment in parameter variation, and the weak perturbations have failed. When confronted with complex texture features, spectral features, and other factors, the limited availability of labeled data makes it challenging for the network to learn comprehensive features, leading to potential classification errors. For example, during the classification of the "building" category, critical feature points incorrectly fall into the "impervious surfaces (ISs)" category (Epoch 10). As the network parameters stabilize and converge toward a local optimum (Epoch 20), the model becomes less responsive to minor data variations, causing cognitive biases to persist. Although the network does learn some features from the "building" category, its primary attention remains focus on the interference area of "ISs." The corresponding prediction results are illustrated in Fig. 1(b) as y_O and y_V. As observed in the figure, the predictions with and without the VAT remain on the same side of the decision boundary. In this case, the network's learning and cognitive bias significantly disrupt the training of semi-supervised semantic segmentation networks for remote sensing images, making it challenging for the network to accurately determine the real decision boundary.

To address these challenges, we propose a bias correction framework (BCF) for semi-supervised semantic segmentation of remote sensing images that effectively alleviates learning and cognitive biases. First, we introduce a feature consistency enhancement module (CEM) to improve the feature representation. Given that the teacher model generally exhibits superior performance and generalization ability [22], [23], the CEM is applied at the feature encoder stage to enable the student model to learn the feature representations of the teacher model. This allows that the student model to be guided by both the label information and the teacher model during the training process, thereby reducing the model's reliance on labeled data and mitigating its tendency to overfit local features. Furthermore, we propose a multidecoder module (MDM) to enhance the feature perturbation, which maximizes feature differences by minimizing the correlation between two corresponding decoder features. As shown in Fig. 1(a), although decoder1 does not focus on the correct "building" class, the weak orthogonality constraint allows decoder 2 to learn features that are as different as possible from those of decoder1, enabling the network to correctly focus on "building." In Fig. 1(b), due to the weak orthogonal relationship between the parameters of decoder1 and decoder2, they make different predictions for the same pixel: the pseudolabel y_2 from decoder2 correctly identifies the "building" class, while decoder1 incorrectly predicts y_1 as "ISs." As the penalty is imposed on the model in step (2), the decision boundary shifts toward the true distribution along the red arrow, alleviating learning and cognitive biases and enhancing the model's generalization ability. Finally, we introduce a symmetrical knowledge transfer mechanism to transfer the diverse, the multiangle information learned by the student model to the teacher model. Different decoders learn features representing various aspects or perspectives of the data. By transferring this diverse information, the teacher model can gain richer feature representations,

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thereby enhancing its generalization ability. To ensure effective symmetrical knowledge transfer, the teacher model employs the same multidecoder structure as the student model. This symmetrical design increases the confidence in pseudolabels and improves the effectiveness of consistency learning, thereby enhancing the overall performance of the model. Specifically, the main contributions of this article are as follows.

1) To address various bias issues in the current semantic segmentation of remote sensing images, we propose a semantic segmentation framework that is more suitable for remote sensing images.
2) We enhance the student model's feature consistency, strengthen the teacher model's guidance to the student model, reduce the risk of overfitting, and improve the quality of feature extraction.
3) An MDM is introduced as a new feature perturbation approach to alleviate the negative impact of network learning bias and cognitive bias during training.
4) We introduce a teacher model structure with symmetric knowledge transfer to improve the confidence of pseudolabels.

The rest of this article is organized as follows. Section II summarizes the related research on semi-supervised semantic segmentation. Section III introduces the proposed BCF in detail, including the network architecture, feature CEM, MDM, and the update of teacher model parameters. Section IV presents ablation experiments and provides quantitative and qualitative comparisons of the proposed method with existing methods to verify the effectiveness of the specific design. Conclusion remarks are provided in Section V.

II. Related Works

The current mainstream semi-supervised semantic segmentation methods can be broadly categorized into two groups: self-training methods and consistency-based semantic segmentation methods [15], [16]. This article provides an overview of these two approaches.

A. Self-Training

The core of the self-training method lies in training a model with labeled data and subsequently generating pseudolabels for unlabeled data [24]. Mittal et al. [25] introduced a method to fully utilize limited pixel-level annotations and additional unlabeled images through adversarial training with a feature-matching loss. Yang et al. [17] proposed an advanced self-training framework called ST++ that prioritizes reliable unlabeled images based on the prediction-level stability. In addition, He et al. [26] proposed a method called distribution alignment and random sampling (DARS) to alleviate biased pseudolabels in semi-supervised semantic segmentation. Ke et al. [27] developed a three-stage self-training framework for semi-supervised semantic segmentation and demonstrated state-of-the-art performance through extensive experiments. Fan et al. [28] introduced the uncertainty guided cross-head co-training (UCC) framework, which combines consistency and self-training via weak and strong augmentation within a shared encoder. Kwon and Kwak [29] proposed an error localization network for semi-supervised semantic segmentation, addressing the problem of cognitive bias in pseudolabeling. Ju et al. [30] introduced a class adaptation framework for semi-supervised semantic segmentation (CAFS) to address the limitations of existing high-confidence-based pseudolabeling methods.

The aforementioned studies collectively underscore the significance of enhancing semantic segmentation performance through the utilization of both labeled and unlabeled data. However, a limitation of such methods is that the quality of pseudolabels depends on the current model's performance; if the segmentation result is incorrect, the error will persist and potentially amplify in subsequent training.

B. Consistency-Based Semantic Segmentation

In recent years, most consistency-based methods have been based on the mean teacher [31], which enforces consistency between the predictions of the student network and the teacher network. The primary distinction among current semi-supervised semantic segmentation methods based on consistency regularization lies in how they integrate perturbations into the data, which can be broadly categorized into three groups: data perturbations, feature perturbations, and network perturbations.

1) Data Perturbations: CutOut [32] and CutMix [33] are classic data perturbation methods widely employed in semi-supervised learning. These methods typically utilize rectangular masks on images. CutOut removes the rectangular content marked by the mask during training. Subsequently, a regularization term is employed to ensure consistency between the prediction made on the original image and the prediction made on the modified image. CutMix combines two images using a rectangular mask to create a new image, where the region marked by the mask belongs to one of the original images and the remainder belongs to the other. ClassMix [34] is applied to the mask of the mixed image. The areas marked by the mask correspond to regions in the image belonging to the same class. Therefore, parts that entirely belong to one class are copied to the other image, creating a new enhanced image. In addition to these methods that propose specific data augmentation techniques for segmentation, other methods utilize classical data augmentation techniques, such as cropping, color jittering, or flipping to generate perturbed versions of the original image.

2) Feature Perturbations: This method perturbs the internal features during the training process. Ouali et al. [14] proposed a semi-supervised semantic segmentation method based on cross-consistency training (CCT), which enhances the representation ability of the encoder and improves the segmentation performance by applying different types of perturbations (such as feature perturbations, prediction perturbations, and random perturbations) on the encoder output and enforcing the prediction consistency between the main decoder and multiple auxiliary decoders. Zhou et al. [35] proposed an uncertainty-aware consistency regularization method to

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enhance the model robustness and consistency in cross-domain semantic segmentation tasks through a dynamic weighting scheme and a class feature perturbation (ClassDrop) strategy. Ma et al. [36] proposed a coarse-to-fine domain-adaptive semantic segmentation pipeline that combines image-level alignment with category-level feature distribution regularization. Wang et al. [37] introduced a feature-level consistency learning framework called density-descent feature perturbation (DDFP) for semi-supervised semantic segmentation, focusing on perturbation-invariant training at the feature level. Together, these studies demonstrate the importance of consistency regularization and perturbation techniques in improving the performance of semantic segmentation methods.

3) Network Perturbations: Ke et al. [38] proposed a guided collaborative training (GCT) method that additionally applies network perturbation by employing two segmentation networks with the same structure but different initializations and enforcing consistency between the predictions of the perturbed networks. Chen et al. [12] proposed a new and straightforward network perturbation consistency regularization method called cross-pseudo-supervision. Similar to GCT, although the two networks share the same architecture, they are initialized with different random weights, increasing their disparity. The model encourages the two perturbation networks to produce highly similar predictions for the same input image and utilizes unlabeled data with pseudolabels to expand the training dataset.

C. Semi-Supervised Semantic Segmentation on Remote Sensing Images

With the successful application of semi-supervised semantic segmentation in various fields, some researchers have proposed semi-supervised learning methods for remote sensing images. Hong et al. [39] introduced a semi-supervised hyperspectral dimensionality reduction method called iterative multitask regression (IMR). This method enhances classification and recognition accuracy by jointly considering labeled and unlabeled data and dynamically propagating labels on a learnable graph to optimize pseudolabels iteratively. Building on this, Hong et al. [40] proposed the X-ModalNet network, which improves decision boundaries by gradually refining pseudo labels. Sun et al. [41] developed BAS^{4}Net, a boundary-aware semi-supervised semantic segmentation network that aims to achieve more accurate segmentation results, particularly at object boundaries, without additional annotation workload. Furthermore, Wang et al. [42] introduced RanPaste, a method combining consistency and pseudolabeling for semi-supervised semantic segmentation of remote sensing images. However, these methods have a drawback. As training progresses, general image enhancement can make it challenging for the network to produce different predictions, significantly reducing the efficiency of consistency learning.

The self-training method and the consistency-based method share a common drawback: the model training accuracy is significantly affected by the quality of the pseudolabels. To address this, we designed a CEM that employs a more stable teacher model to guide the training of the student model and uses a symmetric knowledge transfer method to unbiasedly transfer information learned by the student model back to the teacher model. This dual approach ensures the quality of pseudolabels and improves model segmentation performance. In addition, in consistency-based methods, as training progresses, general feature perturbation can hinder the network's ability to make varied predictions, significantly reducing the efficiency of consistency learning. To mitigate this, the MDM proposed in this article enables different decoders to learn distinct features, effectively alleviating the aforementioned issues.

III. Methods

A. Network Architecture

The overall process is illustrated in Fig. 2. Both labeled data $D_L = \{x_i^l, \text{label}_i\}_{i=1}^{|D_L|}$ and unlabeled data $D_U = \{x_i^u\}_{i=1}^{|D_U|}$ participate in network training. Before training, we apply strong and weak augmentations to the unlabeled data to enhance the model's generalization ability. Strong augmentation (color transformation) is applied to the student model, and weak augmentation (rotation and scaling) is applied to the teacher model. The architecture comprises three components: the encoder, the feature CEM, and the weak orthogonal-constrained multidecoder. Before the training process, the teacher model and the student model are randomly initialized. $E(\theta_T)$ and $E(\theta_S)$ represent the feature encoder of the teacher model and student model, while F_T and F_S represent the highest level feature maps of the teacher model and student model, respectively. The CEM constrains the features learned by the student model, ensuring that they closely resemble those learned by the teacher model. In the decoder stage, the teacher and the student models are designed with a multidecoder structure, where MDM_{T_i} and MDM_{S_i} represent the ith ($i = [1, n]$) MDM of the teacher model and the student model. Here, weak orthogonality constraints are imposed on the MDM of the student model, see Section III-C for details.

B. Feature CEM

The feature CEM is the foundation of our framework, serving two main functions: enhancing feature representation to mitigate the impact of low-quality features on perturbations, and alleviating learning bias. The quality of features extracted by the encoder directly impacts the effectiveness of semantic segmentation in remote sensing images. Given the limited labeled data available in semi-supervised semantic segmentation, the feature encoder struggles to extract comprehensive features. Therefore, we aim to establish constraints enabling the student model to receive guidance from both labeled data and the teacher model. This design is based on two key observations: on the one hand, the teacher model is typically updated using the exponential moving average (EMA) method, resulting in smoother and more stable parameter changes; on the other hand, the teacher model generally represents the average parameters over multiple time steps, integrating information from different time steps and thereby providing a richer feature representation.

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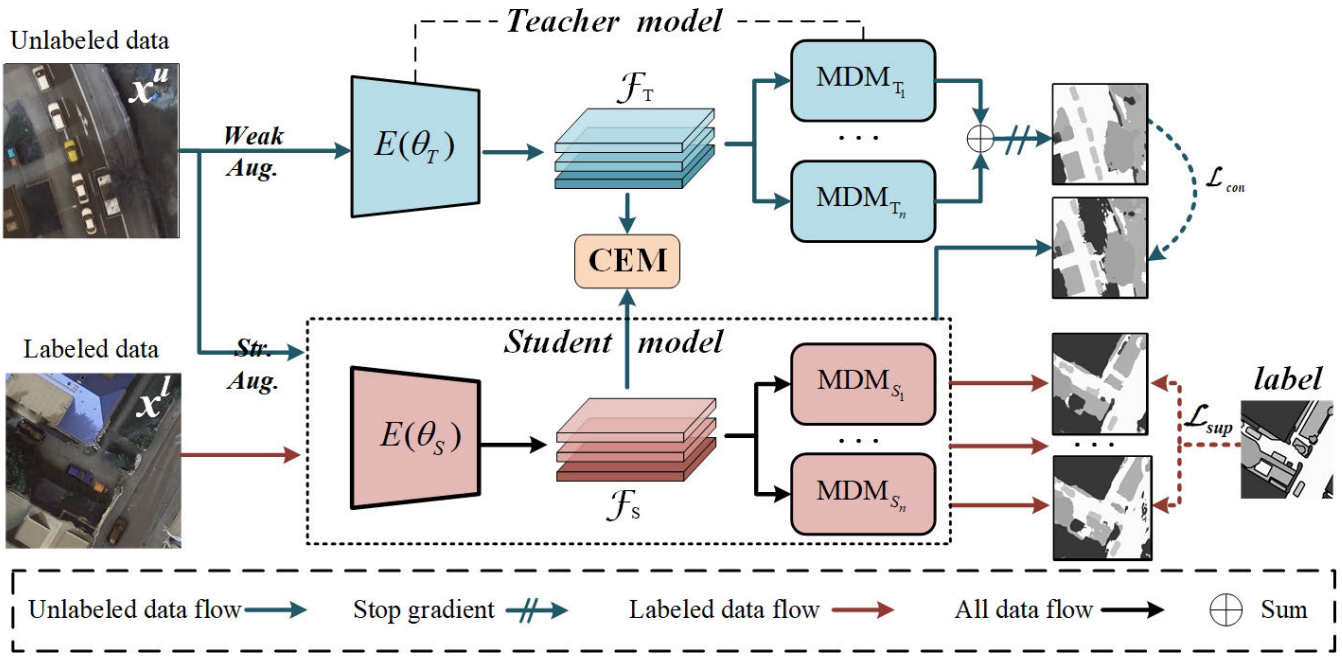


Fig. 2. Architecture of our proposed network. The light blue branches represent the teacher model, and the light orange branches represent the student model. The unlabeled image x^u is weakly augmented for the teacher model. The same unlabeled image is strongly augmented for the student model. The features extracted by the teacher model and the student model from unlabeled data are constrained by CEM. Labeled data x^l is used to supervise each decoder of the student model.

Specifically, during the training process, we introduce a distance constraint that calculates the distance between the features extracted by the teacher model and those extracted by the student model. By minimizing this distance, the student model is encouraged to learn similar feature representations to the teacher model, thereby reducing the risk of overfitting on labeled data and improving the model's generalization ability. This method not only improves the feature representation quality of the student model but also enhances the overall semantic segmentation performance. To achieve this goal, we use the average Euclidean distance in the feature space as the feature loss function. For a given sample, in the feature space of the teacher model and the student model, the Euclidean distance between their feature is calculated, and the average of the distances of all samples is taken as the value of the loss function. By minimizing this loss function, the features of the student model can gradually approach the features of the teacher model, thereby achieving enhanced consistency of the features. The specific formula is as follows:

$$d(F_S, F_T) = \frac{1}{N}\sum_{i=1}^{N}\left|\left|F_S^i, F_T^i\right|\right| \tag{1}$$

where F_S and F_T represent the features of the student model and the teacher model at the last layer, N is the number of features, and $||\cdot||$ represents the Euclidean distance.

C. Multidecoder Weak Orthogonality Constraint Module

As mentioned above, weak perturbations may fail in the middle and later stages of training when dealing with complex remote sensing images. Inspired by previous work [43], unlike previous weak perturbations, we introduce stronger perturbations to the network, i.e., weakly orthogonalize different decoder weights. Our framework consists of multiple decoders that share the same input. Initially, these decoders are assigned different weights through random initialization. For labeled data, each decoder is supervised separately. For unlabeled data, the weighted average of the decoders' prediction results is used as the final output. During the training process, every two decoders of the student model learn different features in a weakly orthogonal manner.

Why do orthogonal weights enable the network to learn different features? In deep neural networks, weights represent the relationship transformation between input and output features, as shown in Fig. 3. Therefore, to minimize the correlation between the features of the two decoders, we maximize the difference in their weights. However, to maintain network stability and leverage the complementarity between multiple decoders, we do not enforce strict orthogonality between the two decoders. Instead, we adopt a more flexible approach with weak orthogonality. In other words, we define the orthogonal function of the two decoders' parameters as the loss function. During training, this loss function is minimized through continuous iterations to ensure that the second decoder learns different feature representations from the first decoder, rather than completely orthogonal ones. This approach enhances the model's performance and generalization ability on complex remote sensing images. The specific loss function is shown in the following formula:

$$\mathcal{L}_{\text{orth}} = \lambda \cdot \left|W \times W^T - I\right|^2, \quad \lambda = \sqrt{1 - \left(\frac{\text{iter}}{\text{total}_{\text{iter}}}\right)}. \tag{2}$$

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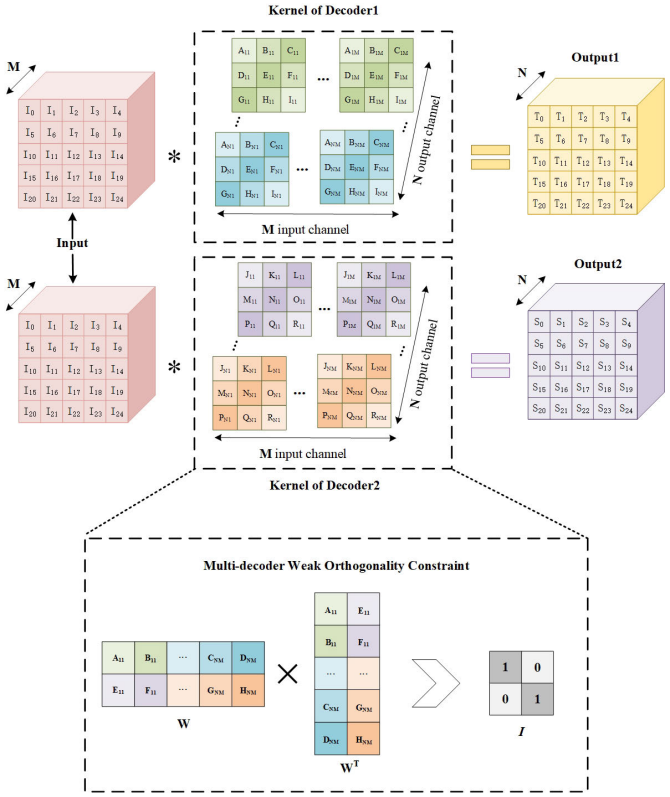


Fig. 3. Schematic of the multidecoder weak orthogonality module. All decoders share the same input, and padding = 1 is used during convolutional operations to ensure that the output size remains unchanged. W is the concatenation of the flattened weights of the two decoders, with the product of W and its transposed W^T as a constraint to make it infinitely close to the identity matrix.

Here, $\mathcal{L}_{\text{orth}}$ represents the regularization loss, λ is a dynamic parameter used to adjust the weight of $\mathcal{L}_{\text{orth}}$, W is the result of flattening the weights of the two decoders and concatenating them together, as shown in Fig. 3, and I represents the identity matrix. Following the gradient descent method, $\mathcal{L}_{\text{orth}}$ approaches 0 as training progresses, implying that $W \times W^T$ converges to the identity matrix. By definition of the identity matrix, decoders 1 and 2 become orthogonal, enabling them to learn distinct features.

It is important to note that, in semantic segmentation of remote sensing images, acquiring more contextual information does not necessarily lead to better segmentation performance. This is because the background in remote sensing images is highly complex, with each pixel being meaningful. Increased contextual information often introduces more interference features. Therefore, to prevent different MDMs from unnecessarily expanding the network's learning scope, we do not impose constraints on them, which means that the areas learned by different MDMs may have a certain degree of overlap.

D. Update of Teacher Model Parameters

Different features represent various aspects or perspectives of the data. By transferring the diverse information learned from the student model's different decoders, the teacher model can acquire richer feature representations. Therefore, we design a multidecoder teacher model based on knowledge transfer symmetry to improve the confidence of pseudo labels. Specifically, the structure of the teacher model is identical to that of the student model, with both adopting a multidecoder architecture with an equal number of decoders. The parameters of the teacher model are not updated through direct training but are indirectly updated using the EMA method based on the student model's parameters. To ensure symmetric knowledge transfer, the parameters of each decoder in the teacher model are updated in correspondence with the parameters of the same decoder in the student model. The EMA update formula is as follows:

$$\theta_t = \theta_{t-1} + (1-\alpha)\theta_s. \tag{3}$$

In this article, θ_t represents the parameters of the teacher model at time step t, θ_{t-1} represents the parameters of the teacher model at time step $t-1$, and θ_s represents the current parameters of the student model. α is a smoothing coefficient between 0 and 1. This smoothing coefficient controls the weight ratio of the old and new parameters. It is usually set to a value close to 1 to ensure the stability of the teacher model parameter update.

This approach enables the teacher model to continuously learn and accumulate knowledge from the student model without being unduly affected by noise in the parameter updates of the student model. This parameter update method can enhance the robustness of the teacher model, enabling it to guide the student model learning process more effectively.

E. Loss Function

In the semi-supervised semantic segmentation framework of this article, the loss function of the student model is as follows:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \beta \cdot \mathcal{L}_{\text{con}} + \gamma \cdot d(F_s, F_T) + \mathcal{L}_{\text{orth}}. \tag{4}$$

In this formula, $\mathcal{L}_{\text{sup}}$ refers to the supervision loss, which mainly uses labeled data to supervise the student model. $\mathcal{L}_{\text{con}}$ represents the consistency loss function, which ensures the model's stability and consistency when predicting unlabeled data. $d(F_s, F_T)$ and $\mathcal{L}_{\text{orth}}$ represent the feature enhancement loss and weak orthogonal loss in Sections III-B and III-C, respectively. β and γ, respectively, represent the weight of the corresponding loss.

$\mathcal{L}_{\text{sup}}$ is calculated using the multiclass cross-entropy loss, defined as follows:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{M}\sum_{i=1}\sum_{c=1}^{n} y_{\text{ic}} \log(p_{\text{ic}}). \tag{5}$$

The basic idea of consistency loss is that the model's output for the same unlabeled input data under different perturbations should be consistent. This consistency constraint can reduce model overfitting and improve the utilization efficiency of unlabeled data. The specific form of the loss function is as follows:

$$\mathcal{L}_{\text{con}}(D_u, \theta_s) = -\frac{1}{|D_U||\Omega|}\sum_{x\in D_u}\sum_{\omega\in\Omega} c(\omega)\mathcal{L}\big(\tilde{y}(\omega), p\theta_s(x)(\omega)\big). \tag{6}$$

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Here, ω denotes the pixel address of the output lattice Ω of the segmentation map, $\widetilde{y}(\omega) \in \{0, 1\}^Y$ represents the segmentation prediction from the teacher model at ω, $p\theta_s(x)(\omega) \in [0, 1]^Y$ is the segmentation prediction from the student model at ω, and $c(\omega) \in [0, 1]$ indicates the segmentation prediction confidence from the teacher model at ω.

IV. Experimental Results

A. Dataset

The ISPRS Potsdam dataset consists of 38 aerial images with a spatial resolution of 5 cm. Each image contains four channels: near-infrared (NIR), red (R), green (G), and blue, with a size of 6000 × 6000 pixels. The dataset is labeled according to six land cover types: ISs, buildings, low vegetation (LV), trees, cars, and background. The division of the training set, test set, and verification set in this article is consistent with [44], 18 images are used for training, six images are used for verification, and the remaining 14 images are used for testing. To ensure that the labeled and unlabeled data share the same distribution and do not overlap, we first crop each image into 3000 × 3000 blocks and then randomly divide them into labeled and unlabeled datasets in the ratio of 1:7 and 1:15.

The ISPRS Vaihingen dataset is also a high-resolution aerial image dataset, which includes 33 images of different sizes. Each image has a resolution of 9 cm and includes three bands: NIR, R, and G. Similar to Potsdam, it includes pixel-level annotations for six types of features: ISs, buildings, LV, trees, cars, and background. We select three images for validation (IDs: 11, 28, and 34) and two images for testing (IDs: 15 and 30), and use the remaining images for training. Each training image is cropped into four nonoverlapping images, which are then divided into corresponding labeled and unlabeled data according to a specified ratio (1:7 and 1:15).

B. Implementation Details

To evaluate the performance of our method, we add a student decoder based on the mean teacher model and use it as the baseline network for this study. In the encoder stage, the feature layer output by each block has the following number of channels: [64, 256, 512, 1024, 2048]. After multiscale feature processing, the channel dimension is reduced to 256. In the decoder, the resolution is restored using convolution and upsampling, with the final output channels corresponding to the number of categories for segmentation (as shown in Fig. 4). The encoder and decoder of both the teacher and student models follow this structure. To fully utilize computing resources and effectively expand the dataset, a 512 × 512 window is applied to randomly crop input images during training, generating training samples. Specifically, for each input image, a random top-left corner is generated, and a 512 × 512 crop is extracted based on this position. As described in [45] and [44], the number of crops per image block is set to 100. During the inference stage, input image patches are cropped from the original test image using the sliding method and processed individually. To evaluate the performance of our method, we utilize three evaluation metrics: mean intersection over union (mIOU), $F1$ score, and overall accuracy (OA).

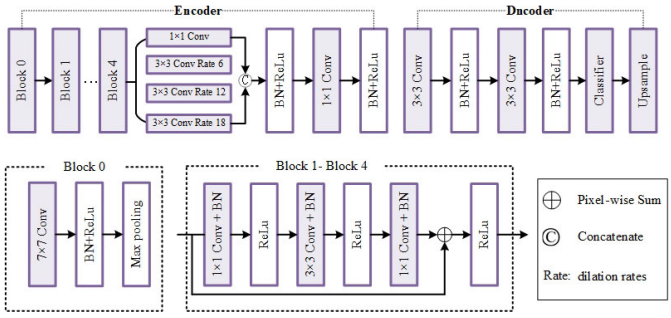


Fig. 4. Architecture of encoder and decoder.

The mIOU computes the average intersection-over-union for each class in the dataset

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{7}$$

$$\text{mIOU} = \frac{\sum_{i=1}^{n} \text{IOU}_i}{n}. \tag{8}$$

The $F1$ score is the harmonic mean of precision and recall

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{10}$$

OA considers the overall classification accuracy. These metrics are calculated using the counts of true positives (TPs), true negatives (TN), false positives (FP), and false negatives (FN) pixels as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{11}$$

The clustering learning method employs a stochastic gradient descent optimizer with specific settings: a base learning rate of 0.01, momentum of 0.9, and weight decay of 0.00001. The training process spans 40 epochs, with the initial learning rate set to 0.01. After each iteration, the learning rate is multiplied by $[1 - (\text{iter}/\text{total}_{\text{iter}})]^{0.9}$. The momentum decay coefficient is also set to 0.9, and the weight decay coefficient is set to 0.0001. The netswork was developed using PyTorch and trained on a TITAN RTX × 24 GB GPU.

C. Ablation Study

Ablation experiments are performed on the Potsdam dataset and the Vaihingen dataset of ISPRS. Since the method in this article primarily addresses scenarios where unlabeled data significantly exceeds labeled data, we select 1/8 and 1/16 of the labeled data for experimental verification. Then, we add consistency enhancement and a multidecoder structure to the baseline model for experimental verification. Among them, CEM stands for the feature CEM, while MDM denotes the multidecoder weak orthogonality constraint module. The number preceding MDM indicates the number of MDMs. In the table, "+CEM" signifies the addition of only the CEM to the baseline. The notation "+2 MDM" indicates the incorporation

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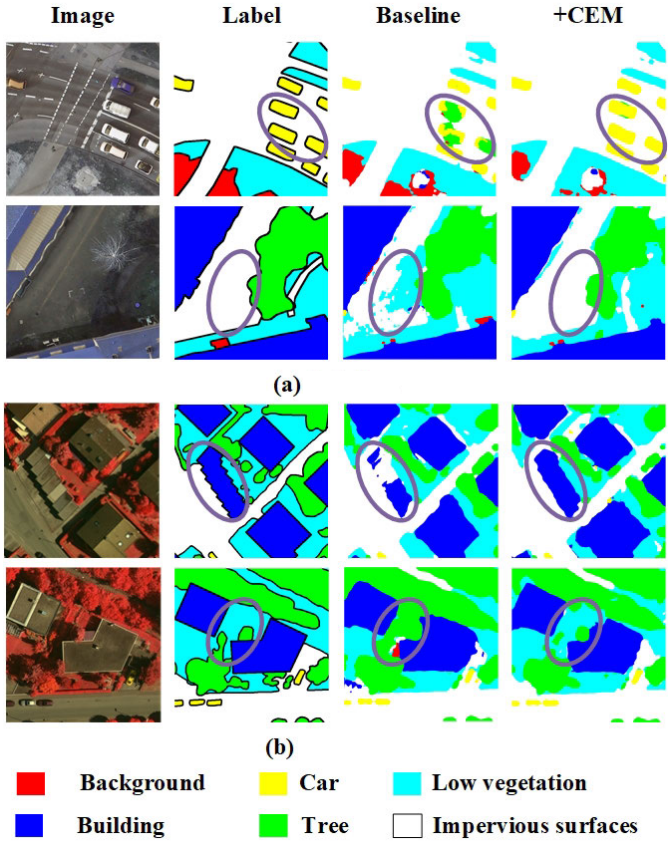


Fig. 5. Comparison of prediction results among baseline and our method. (a) Potsdam. (b) Vaihingen.

TABLE I
ABLATION EXPERIMENTS ON POTSDAM DATASET

Labeled data	ModelName	Evalution index		
		OA	mIOU	Avg.F1
1/8	Baseline	85.86	78.89	87.96
	+CEM	86.04	79.49	88.32
	+2 MDM	**86.18**	**79.37**	**88.25**
	+CEM +2 MDM	86.09	78.99	88.03
1/16	Baseline	83.24	74.91	85.39
	+CEM	83.92	76.15	86.12
	+2 MDM	84.51	77.83	87.3
	+CEM +2 MDM	**85.54**	**78.53**	**87.7**

TABLE II
ABLATION EXPERIMENTS ON VAIHINGEN DATASET

Labeled data	ModelName	Evalution index		
		OA	mIOU	Avg.F1
1/8	Baseline	87.04	77.42	87.07
	+CEM	87.34	78.00	87.42
	+2 MDM	87.58	78.68	87.87
	+CEM +2 MDM	**88.36**	**80.04**	**88.73**
1/16	Baseline	85.85	74.36	85.04
	+CEM	86.59	75.35	85.72
	+2 MDM	87.4	77.64	87.19
	+CEM +2 MDM	**88.07**	**78.01**	**87.91**

of two MDM modules into the baseline. Finally, "+CEM + 2 MDM" refers to the simultaneous introduction of the CEM module and two MDM modules into the network.

1) Effects of the Feature CEM: As shown in Tables I and II, the segmentation results after adding the CEM are compared with the baseline. Compared with the Baseline method, significant improvements are observed when labeled data accounts for 1/8 and 1/16 of the total data volume in both the Potsdam and Vaihingen datasets. Experiments have demonstrated that enhancing the guidance provided by the teacher model to the student model can assist the latter in extracting features with enhanced generalization capabilities, consequently improving the performance of semi-supervised semantic segmentation. Fig. 5 illustrates the case where the labeled data ratio is 1/16, comparing prediction results before and after incorporating CEM. The figure shows that improving feature quality can effectively alleviate learning bias. Meanwhile, using the results from the Vaihingen dataset as a case study, the addition of the CEM when the proportion of labeled data is reduced to 1/8 and 1/16 leads to an improvement in overall segmentation accuracy by 0.3% and 0.74%, respectively. A comparison of these results reveals that as the proportion of labeled data decreases, the CEM proposed in this study becomes increasingly effective (similar conclusions are drawn from the Potsdam dataset). This phenomenon occurs because, with a limited amount of labeled data, the model tends to overfit these data. This further validates that the method proposed in this article can enhance feature extraction quality and effectively reduce the model's reliance on labeled data.

2) Effects of MDM : We will analyze the impact of the number of decoders on the segmentation performance in the next section. Here, we take two MDMs as an example [(i.e., four decoders)]. As shown in Tables I and II, when the labeled data is 1/8 and 1/16, the Potsdam dataset and the Vaihingen dataset have obvious improvements in OA, mIOU, and average $F1$. Similar to the results of CEM, the overall improvement is much greater when the proportion of labeled data is 1/16 compared to when it is 1/8. This proves that the decoders' weakly orthogonal method can also effectively alleviate the model's over-reliance on labeled data. Comparing Tables I and II, we observe that, unlike the experimental results of CEM, the overall improvement of the Vaihingen dataset is much greater than that of the Potsdam dataset. This is because the overall data volume of the Vaihingen dataset is much smaller than that of Potsdam. In semi-supervised semantic segmentation, unlabeled data is used to enhance

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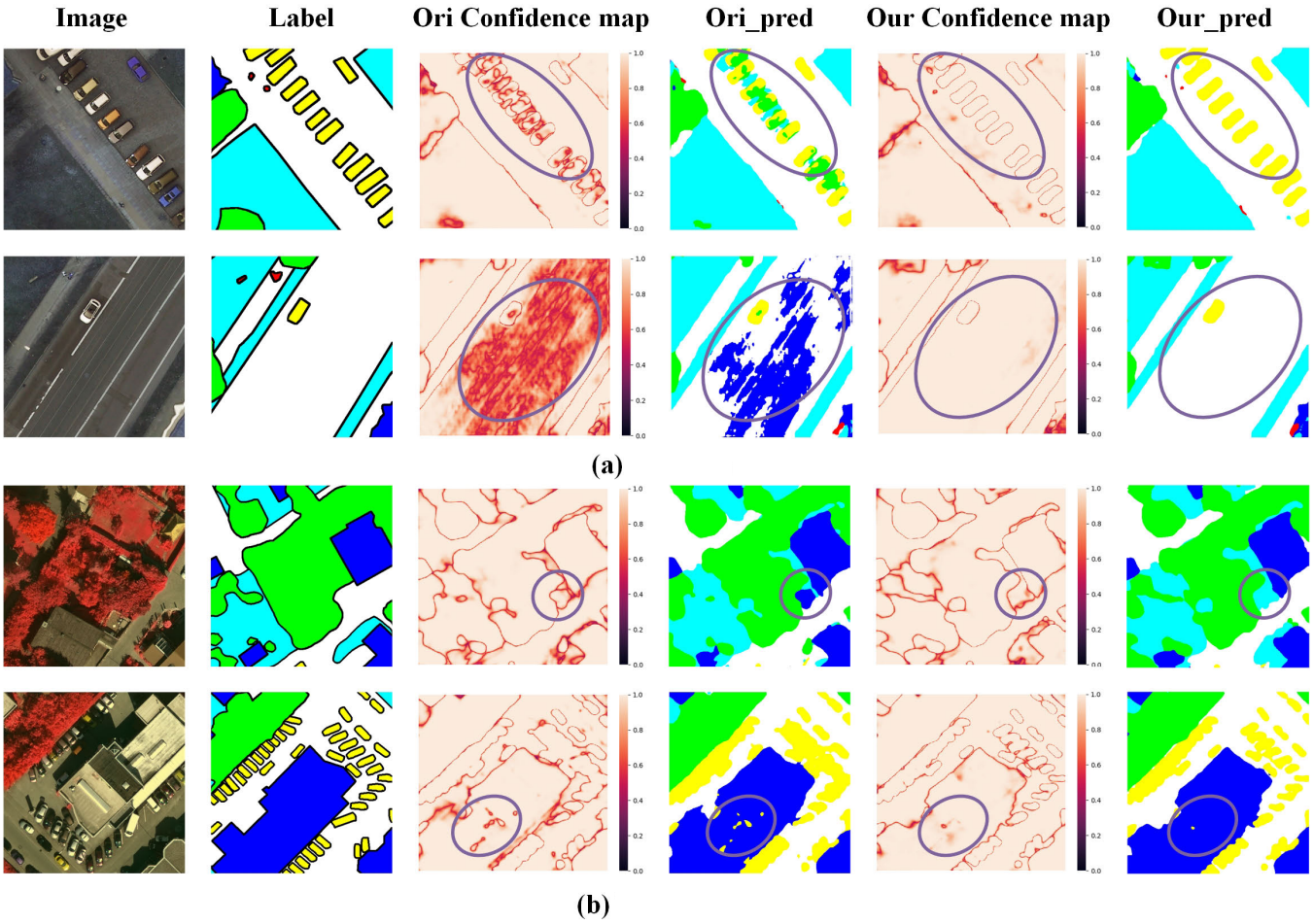


Fig. 6. Comparison of prediction results between baseline and our method. (a) Potsdam. (b) Vaihingen.

the training set through pseudolabel generation. Suppose the amount of labeled data is small. In that case, the performance of the initial model may be poor, resulting in low-quality generated pseudolabels, which in turn affects the subsequent training of the model. Hence, the method presented in this article effectively mitigates learning and cognitive biases, enhances the confidence of pseudolabels, and thereby ensures the efficacy of consistency regularization.

3) Combined Effects of CEM and MDM: The bold fonts in Tables I and II indicate the best results in the experiments. After combining consistency enhancement and weak orthogonality, we observe that both the Potsdam dataset and the Vaihingen dataset achieve their best results when the data size is 1/16. To better verify the effect of this method on cognitive bias and learning bias, as shown in Fig. 6, we use the trained Baseline model and our model to make predictions and calculate the corresponding confidence. By comparing the third and fourth columns of the first row, we observe that the initial model is influenced by learning bias, the network mistakenly predicts "car" as "tree," even though there are large feature differences between these categories. It can be seen from the third and fourth columns of the second row that, affected by cognitive bias, the Baseline network often confuses "IS" with "building." The same problem also exists on the Vaihingen dataset. A similar issue is observed in the corresponding Vaihingen dataset, where the network incorrectly predicts "building" as "car" and confuses "LV" with "building." The fifth and sixth columns display our method's confidence and prediction results. Comparing the fifth and third columns, we observe that our method can effectively mitigate the effects of learning bias and cognitive bias, enabling the network to focus on the correct areas. Comparing the prediction results in the fourth and sixth columns, we can more intuitively see the effectiveness of our method. However, when the data volume was reduced to 1/8, the Potsdam dataset did not yield the best results. Compared to using only weak orthogonality, OA decreased by 0.09%, mIoU by 0.38%, and the average $F1$ score by 0.22%. Although these declines are small, the results are not as strong as in other experiments. Upon analysis, both modules proposed in this article aim to alleviate learning and cognitive biases, and their functions are not entirely independent. When the feature distributions of labeled and unlabeled data are similar, fewer interfering features appear in the unlabeled data. In this scenario, MDM alone provides sufficient diversity, and further feature enhancement may add complexity to the model without significant performance gains. In this case, the additional enhancement may not improve feature distinction

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TABLE III
PARAMETERS OF DIFFERENT DECODERS

Data	ModelName	Evalution index		
		Parameters/M	Flops/GMac	FPS
Potsdam	Baseline	41.76	79.33	86.06
	+ 2 decoder	43.05	100.53	74.47
	+ 3 decoder	44.34	121.73	65.22
	+ 4 decoder	45.63	142.93	57.22
	+ 5 decoder	46.92	164.13	51.98

TABLE IV
EFFECT OF THE NUMBER OF DECODERS ON THE POTSDAM DATASET

Labeled data	ModelName	Lorth	Evalution index		
			OA	mIOU	Avg.F1
1/8	Baseline	W/O	85.86	78.89	87.96
	2 decoder	W/O	85.78	78.9	87.94
		W/	**86.32**	**79.52**	**88.39**
	3 decoder	W/O	85.18	78.02	87.39
		W/	85.73	78.94	87.99
	4 decoder	W/O	85.62	78.63	87.82
		W/	86.18	79.37	88.25
	5 decoder	W/O	85.44	78.69	87.58
		W/	86.21	79.49	88.32
1/16	Baseline	W/O	83.24	74.91	85.39
	2 decoder	W/O	84.05	76.68	86.57
		W/	84.35	77.11	86.78
	3 decoder	W/O	84.5	77.31	87.00
		W/	85.24	78.32	87.58
	4 decoder	W/O	83.67	76.05	86.12
		W/	84.51	77.83	87.3
	5 decoder	W/O	84.57	77.89	87.16
		W/	**85.4**	**78.6**	**87.73**

meaningfully and could even introduce minor noise, leading to potential misjudgments by the model.

4) Parameters of Different Decoders: To evaluate the impact of the number of decoders on the entire inference process, we use as an example. We assess the parameters, floating point operations per second (FLOPs), and frames per second (FPS) under different numbers of decoders. From Table III, we observe that the number of model parameters, FLOPs, and FPS increase significantly as the number of decoders increases. This indicates that although increasing the number of decoders may enhance the semantic segmentation performance of the model, it also increases the model's complexity and computational requirements, potentially leading to longer training and inference times. Therefore, there is a tradeoff between performance improvements and computational overhead when designing and optimizing models.

TABLE V
EFFECT OF THE NUMBER OF DECODERS ON THE VAIHINGEN DATASET

Labeled data	ModelName	Lorth	Evalution index		
			OA	mIOU	Avg.F1
1/8	Baseline	W/O	87.04	77.42	87.07
	2 decoder	W/O	87.74	78.95	87.92
		W/	**88.19**	**79.74**	**88.54**
	3 decoder	W/O	87.82	79.39	88.28
		W/	88.25	79.77	88.61
	4 decoder	W/O	86.87	76.58	86.50
		W/	87.58	78.68	87.87
	5 decoder	W/O	87.67	78.16	87.51
		W/	**88.58**	**80.19**	**88.81**
1/16	Baseline	W/O	85.85	74.36	85.04
	2 decoder	W/O	86.52	75.32	86.38
		W/	87.02	76.24	87.3
	3 decoder	W/O	86.99	76.73	86.64
		W/	87.28	77.58	87.2
	4 decoder	W/O	86.22	75.22	85.63
		W/	87.4	77.64	87.19
	5 decoder	W/O	86.22	75.3	85.73
		W/	**87.89**	**78.39**	**87.71**

5) Effect of the Number of Decoders: To further investigate the relationship between the weak orthogonal loss ($\mathcal{L}_{\text{orth}}$) and the number of decoders, we conducted ablation experiments to evaluate the effects of different numbers of decoders with and without $\mathcal{L}_{\text{orth}}$. "W/O" denotes the absence of $\mathcal{L}_{\text{orth}}$, while "W/" indicates its inclusion. When the number of decoders is even, weak orthogonality is applied between each pair of decoders. In contrast, for an odd number of decoders, one decoder does not undergo orthogonalization with any other decoder. The experimental results are shown in Tables IV and V. The results indicate that for a given number of decoders, models without $\mathcal{L}_{\text{orth}}$ perform significantly worse than those with it. In addition, increasing the number of decoders does not consistently improve results. Without $\mathcal{L}_{\text{orth}}$, the model's performance does not improve as more decoders are added. Meanwhile, after incorporating $\mathcal{L}_{\text{orth}}$, Table IV shows that, with only 1/8 of the data labeled, the model achieves optimal performance with two decoders. There are two reasons for this phenomenon: first, too many decoders may generate redundant observations. As the number of decoders increases, the network finds it hard to learn more effective features and redundant observations interfere with the final training results. Second, increasing the number of decoders raises the number of model parameters and computational complexity. An overly complex model may lead to overfitting. Taking into account the training speed and model stability, this article selects four decoders (i.e., two MDMs) for the experiments.

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TABLE VI
COMPARATIVE EXPERIMENT WITH POTSDAM DATASET

Labeled data	Model	F1 score per category(%)					Evalution index		
		IS	building	LV	Tree	Car	OA	mIOU	Avg.F1
1/8	Baseline	89.22	93.15	80.39	83.45	93.58	85.86	78.89	87.96
	S4GAN	88.64	92.01	**81.36**	81.93	84.09	**86.49**	75.62	85.61
	MT	88.03	92.84	80.01	80.96	88.63	84.49	75.92	86.09
	CCT	87.69	91.74	81.7	82.69	83.87	85.22	74.92	85.54
	PS-MT	89.16	93.35	80.84	**84.79**	**93.59**	86.11	79.47	88.35
	RCL	88.65	93.31	81.3	82.59	89.49	85.57	77.33	87.07
	CACC	-	-	-	-	-	-	77.33	87.04
	Ours	**89.43**	**93.63**	81.24	84.66	92.99	86.32	**79.52**	**88.39**
1/16	Baseline	86.17	91.49	77.42	80.98	90.9	83.24	74.91	85.39
	S4GAN	87.49	91.65	**80.48**	80.37	83.26	85.37	74.23	84.65
	MT	85.96	91.19	76.96	78.99	87.33	82.27	72.95	84.09
	CCT	84.88	88.88	79.52	79.02	81.98	82.38	70.91	82.86
	PS-MT	84.82	91.99	77.04	81.61	92.64	82.62	75.34	85.62
	RCL	88.02	93.31	81.30	82.59	89.49	84.36	76.03	86.14
	CACC	-	-	-	-	-	-	75.07	85.61
	Ours	**88.85**	**93.53**	79.8	**82.88**	**93.44**	**85.54**	**78.53**	**87.7**

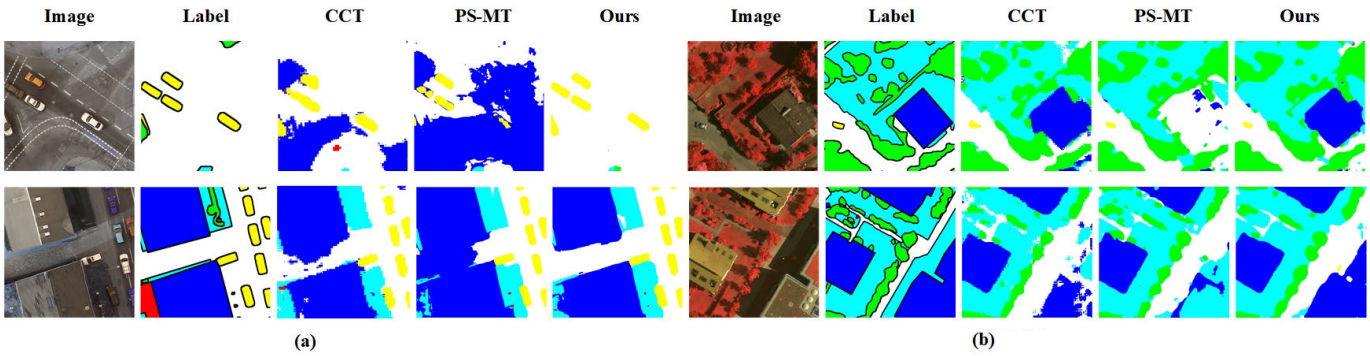


Fig. 7. Comparison of prediction results among CCT, PS-MT, and our method. (a) Potsdam. (b) Vaihingen.

D. Comparison With Previous Methods

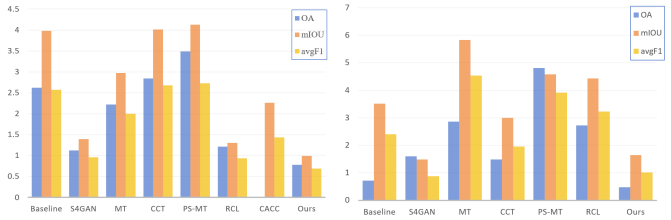
To demonstrate the superiority of our approach, this section compares our method with several state-of-the-art semi-supervised semantic segmentation methods. Including S4GAN [25], MT [31], CCT [14], PS-MT [19], RCL [45], and CACC [46]. Some of these experimental results are based on the results in this article, while others are retrained based on the source code. The results of S4GAN refer to the results of paper [47]; both MT and RCL results are from [45]; the results of CACC use the results of [46]. CCT and PS-MT are the results of retraining based on the source code.

1) Results on the Potsdam Dataset: Table VI lists the segmentation results of each method on the Potsdam dataset. When the labeled data accounts for 1/8 of the total data, our method achieves 86.32% on OA, 79.52% on mIOU, and 88.39% on average $F1$. Our method achieves the best results for both mIOU and average $F1$, while OA is slightly lower than the result of S4GAN. After analysis, this difference may be because the S4GAN experiment involves a certain degree of overlap between labeled and unlabeled data, while our two datasets do not overlap at all. When the labeled data accounts for 1/16 of the total data, our method achieves 85.54% on OA, 78.53% on mIOU, and 87.70% on average $F1$, all of which outperform other methods. Fig. 7 shows the comparison of the prediction results of different methods, using CCT and PS-MT as examples. As shown in the figure, despite the application of various weak perturbations by the CCT and PS-MT methods, the network remains affected by learning bias and cognitive bias, leading to large-scale misclassification. Our method effectively alleviates the impact of these biases, producing superior results.

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TABLE VII
COMPARATIVE EXPERIMENT WITH VAIHINGEN DATASET

Labeled data	Model	F1 score per category(%)					Evalution index		
		IS	building	LV	Tree	Car	OA	mIOU	Avg.F1
1/8	Baseline	87.97	94.14	76.89	88.66	87.36	87.04	77.42	87.07
	S4GAN	83.09	84.97	73.31	83.06	60.28	81.16	63.40	76.94
	MT	87.00	90.58	75.75	82.62	74.99	83.94	70.26	82.19
	CCT	88.01	93.03	79.04	88.44	87.66	87.14	77.65	87.24
	PS-MT	74.24	91.98	60.56	87.75	88.79	79.80	69.13	80.66
	RCL	88.30	92.19	78.57	82.77	79.29	85.45	73.14	84.22
	Ours	**89.75**	**94.53**	**80.33**	**89.01**	**90.02**	**88.36**	**80.04**	**88.73**
1/16	Baseline	87.25	92.59	76.34	88.62	78.55	86.32	73.91	84.67
	S4GAN	82.53	84.02	**72.36**	82.88	58.49	79.56	61.92	76.06
	MT	85.82	88.16	68.95	80.37	64.98	81.08	64.43	77.66
	CCT	86.8	91.43	76.86	87.34	84.00	85.66	74.65	85.29
	PS-MT	84.58	90.55	73.29	88.65	85.76	84.61	73.71	84.57
	RCL	87.19	90.31	71.00	81.14	75.29	82.73	68.71	80.99
	Ours	**89.58**	**93.67**	**79.85**	**88.79**	**86.64**	**87.89**	**78.39**	**87.71**



(a) (b)

Fig. 8. Accuracy drop rate of the Potsdam and Vaihingen datasets across different segmentation methods. The overall decrease in OA, mIOU, and Average $F1$ is shown by subtracting the result with 1/16 of the labeled data from the result with 1/8 of the labeled data. (a) Potsdam. (b) Vaihingen.

It is known that as the number of labeled data decreases, the segmentation effect of the network will inevitably decrease to a certain extent, even if the total amount of data remains unchanged. However, compared with other segmentation methods, our method can effectively alleviate this situation. As shown in Fig. 8(a), we measure the decrease in OA when the labeled data is reduced from 1/8 to 1/16. Compared with the other seven methods, our method exhibits the smallest decrease in OA, mIOU, and average $F1$.

2) Results on the Vaihingen Dataset: Table VII presents the segmentation results of each method on the Vaihingen dataset, further demonstrating the effectiveness of the method proposed in this article. The table indicates that when the amount of labeled data is 1/8 and 1/16, our method achieves the best results in terms of OA, mIOU, and average $F1$. In addition, the $F1$ score for each class also achieves the best results. Consistent with the conclusion for the Potsdam dataset, on the Vaihingen dataset, when the number of labeled data is reduced from 1/8 to 1/16, the OA of our method decreases minimally, as shown in Fig. 8(b).

The overall experimental results demonstrate that our method has significantly reduced the model's learning bias and cognitive bias in alleviating semi-supervised semantic segmentation methods. Our method achieves superior performance on various remote sensing image datasets compared with traditional semi-supervised learning methods. These findings indicate that the method proposed in this article can enhance the performance and stability of semi-supervised remote sensing image semantic segmentation.

V. CONCLUSION

This article proposes a BCF to address the problems of learning bias and cognitive bias in semi-supervised semantic segmentation of remote sensing images. The framework improves the quality of feature extraction through a feature CEM. Subsequently, a multidecoder structure is designed to subject the network to significant perturbations using weakly orthogonal weights, compelling the model to learn features from various perspectives. This approach alleviates learning bias and cognitive bias while enhancing the network's generalization ability. Finally, a teacher model structure based on knowledge transfer symmetry is designed to facilitate the transfer of information learned by the student model to the teacher model, enhancing the confidence of pseudo labels and improving the consistency-learning of the network. Finally, we conduct extensive ablation experiments to evaluate the impact of various components of the network framework. Experimental results on the ISPRS Potsdam and Vaihingen datasets demonstrate that the proposed framework significantly outperforms the baseline methods.

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However, this method has some limitations, including the following two points. First, in remote sensing scenarios, the imbalance of class distribution between labeled and unlabeled datasets is inevitable. Models pretrained on insufficiently labeled data exhibit poor generalization ability and tend to produce scattered features for unlabeled data, in which this article does not fully address. Second, as the amount of unlabeled data increases, the issue of catastrophic forgetting becomes more pronounced and remains challenging to address. Future research can focus on these two directions.

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Li Zhang received the M.S. degree from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2021, where she is currently pursuing the Ph.D. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, with a focus on photogrammetry and remote sensing.

Her research interests include semantic segmentation of remote sensing images, deep learning, and image inpainting.



Zhenshan Tan received the B.S. and M.S. degrees from Wuhan University, Wuhan, China, in 2016 and 2019, respectively, and the Ph.D. degree from Fudan University, Shanghai, China, in 2023.

He is currently a Lecturer with the School of Computer Science, Nanjing University of Information Science and Technology, Nanjing, China. His research interests include computer vision, cross-modal retrieval, networking, and information security.



Yuzhi Zheng received the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2020.

She is currently a Research Associate with the School of Law, Wuhan University. Her research interests include cross-cutting areas of remote sensing and law, such as exploring the legal aspects of environmental issues by means of remote sensing.



Guo Zhang received the B.E. and Ph.D. degrees in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2000 and 2005, respectively. His doctoral dissertation was on the rectification of high-resolution remote sensing imaging in the absence of ground control points.

Since 2005, he has been working at the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, where he became a Professor in 2011. His research interests include space photogrammetry, geometric processing of spaceborne optical/synthetic aperture radar (SAR)/interferometric synthetic aperture radar (InSAR) imagery, altimetry, and high-accuracy image matching.



Wen Zhang received the M.S. degree from the Research Center of Graphic Communication, Printing, and Packaging, Wuhan University, Wuhan, China, in 2021.

In 2021, she joined the School of Information Management, Wuhan University to pursue the Ph.D. degree. Her research interests include visual cognition and low-level vision.



Zhijiang Li received the Ph.D. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2005.

He worked with the Research Center of Graphic Communication, Printing, and Packaging and the School of Information Management, Wuhan University, for 18 years, where he became a Professor in 2021. His research interests include visual cognition, natural language processing, and metaverse.

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